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Bitcoin investment: a mixed methods study of investment motivations

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ABSTRACT

Bitcoin is a well-established blockchain-based cryptocurrency that has attracted a great deal of attention from media and regulators alike. While millions of individuals invest in bitcoin, their motivations for doing so are less clear than with traditional investment decisions. We argue that the technical nature of bitcoin investments gives it unique characteristics and, consequently, that we lack a thorough understanding of how this affects the motivations behind bitcoin investment. We use a mixed method approach consisting of qualitative (n = 73) and quantitative (n = 150) studies and fuzzy-set qualitative comparative analysis (fsQCA) to identify seven bitcoin-specific motivations (profit expectancy, ease of bitcoin acquisition, support of bitcoin ideology, investment skills, risk affinity, anticipated and experienced inaction regret) and how configurations of them explain bitcoin investment. The findings reveal, among others, that some individuals invest in bitcoin because they support the bitcoin ideology. Contrary to the traditional investment literature, profit expectancy is not a necessary condition to the extent that there is one empirical configuration of motivations that explains that individuals also invest in bitcoin even if they do not expect profits. The results disclose non-trivial investment motivation configurations and lay the groundwork for future studies of the role of cryptocurrencies in society.

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1. Introduction

It is estimated that around 25 million individuals worldwide possess bitcoin[1] which has a market capitalisation of around 186 USD billion.[2] Those millions of users mostly see their bitcoin engagement as an investment (Lielacher, 2019) and a variety of stakeholders stand to benefit by better understanding what motivates people to invest in bitcoin. Bitcoin companies such as wallet providers or crypto exchanges need to understand why individuals invest in bitcoin in order to optimise their marketing and public relations activities in an effort, among others, to establish bitcoin as a widely accepted medium of exchange and an accepted cryptocurrency (Mattke et al., 2020). Investors' understanding of the dynamics of bitcoin prices rests upon their understanding of underlying investment reasons. Organisations aiming to finance new ventures by launching their own cryptocurrency, e.g., through initial coin offering (ICO) or security token offering (STO) (Fisch, 2019), need to understand why individuals invest in cryptocurrency in order to draw conclusions for the design of their own coins or tokens. In summary, developing a better understanding of what drives investing in bitcoin is relevant for a variety of stakeholders in research and practise.

While there are various theories to explain why individuals invest (e.g., prospect theory or rational choice theory), we rely on regret theory (Loomes & Sugden, 1982) as one of the most used theoretical lenses in investment research. It has been used to understand portfolio investment (Solnik & Zuo, 2012), currency hedging (Michenaud & Solnik, 2008) and IT investment (Lankton & Luft, 2008; EH Park et al., 2016). The central tenet of regret theory is that three distinct motivations influence whether an individual invests (Loomes & Sugden, 1982). First, perceptions about the investment object, the company and its products (Nagy & Obenberger, 1994) motivate investment behaviour. Bitcoin investment research indicates that profit expectancy influences bitcoin investment (Glaser et al., 2014). Second, individual personal factors, such as personal risk attitude (Cohn et al., 1975), motivate investment decisions. Research findings are available in related contexts, but the personal factors driving investment in bitcoin have not yet been identified. Third, regret sentiments influence investments because individuals do not always make rational decisions (Loomes & Sugden, 1982). While this has been tested in other investment contexts, such as stock and equity investment (Jain et al., 2015; Rau, 2015), the influence of regret sentiments on investing in bitcoin remains unknown.

We posit that bitcoin has some unique characteristics that may motivate people to invest in it. In contrast to classic investments, a bitcoin investment

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is not associated with any person, organisation or intermediary, does not finance any venture, and individuals do not acquire a stake in a company or venture. Furthermore, the bitcoin price does not depend on traditional financial assets, such as stocks, and because of its volatility bitcoin does not fulfil the requirements of a currency. Those unique characteristics limit the transfer of knowledge from related research to the context of investing in bitcoin. For these reasons, we apply regret theory with the aim of gaining insights into which motivations, in terms of perception about bitcoin, personal factors and regret sentiments, drive bitcoin investment, asking the following research question (RQ):

RQ1: What perceptions about bitcoin, personal factors and regret sentiments motivate initial bitcoin investment?

While it is initially relevant to know the motivations relevant to the Bitcoin context, complex individual behaviour is driven by configurations of simultaneous coexisting motivations (Liu et al., 2017). Hence, we posit that configurations of simultaneous coexisting perception about bitcoin, personal factors and regret sentiments collectively drive bitcoin investment. In such configurations, the presence (e.g., presence of profit expectancy) or the absence (e.g., absence of profit expectancy) of distinct motivations can be related to the behaviour or distinct motivations can get overridden so that a motivation is not related to the behaviour (e.g., ease of use is a "don't care situation" meaning that it can either be present or absent). The influence of configurations of motivations on behaviour has been applied in Bitcoin research. For example, bitcoin research has shown that the technological complexity of bitcoin causes individuals to behave differently in response to bitcoin, so that there is no one configuration that explains individuals' behaviour. Rather, the different behaviour of individuals is explainable by multiple configurations, where motivations regarding bitcoin can either be present and/or absent (Mattke et al., 2018). This is supported by practical insights from a Canadian study indicating that the motivations for investing in bitcoin differ among individuals (Henry et al., 2018). We therefore ask:

RQ2: What configurations of motivations influence initial bitcoin investment?

To answer both research questions, we apply regret theory (Loomes & Sugden, 1982) and conduct a mixed methods study with two sequential studies (Venkatesh et al., 2013). Study 1 follows a qualitative research design based on 73 interviews to identify the motivations constituting components of configurations of motivations that influence bitcoin investment (RQ1). Overall, we identify seven motivations. We identify profit expectancy, ease of bitcoin acquisition and support of bitcoin ideology as relevant perceptions about bitcoin that drive investing in bitcoin. In addition, we identify investment skills and risk affinity as personal factors that, grouped into configurations, influence bitcoin investment. Finally, we identify anticipated inaction regret and experienced inaction regret as regret sentiments which are components of configurations of motivations that influence bitcoin investment. We use the identified motivations in our quantitative Study 2, in which we apply fuzzy-set qualitative comparative analysis (fsQCA) to survey data (N = 150) to reveal configurations of motivations (RQ2). Based on our findings, we identify four sufficient configurations of motivations for bitcoin investment, which we categorise as four types of potential bitcoin investors: the incompatible profit investor, the risk affine profit investor, the not-profit-driven ideological investor and the experienced ideological investor.

The remainder of this paper is structured as follows. In Section 2, we provide an overview of bitcoin as an exclusively IS-enabled and IS-governed investment and summarise previous research about bitcoin. In Section 3, we outline regret theory and previous research on investment. In Section 4, we describe our manner of answering the two research questions based on a mixed methods research approach, explain our two studies and validate and present the results. Additionally, we draw the meta-inferences from both studies. In Section 5 we discuss the theoretical and practical contributions and we conclude in Section 6.

2. Research context

In this section, we explain the basics of Bitcoin blockchain application and show how investing in bitcoin is exclusively IS-enabled and IS-governed and thus relevant to IS research. We also summarise the current state of bitcoin research.

2.1. Bitcoin as an exclusively IS-enabled and IS-governed investment

The Bitcoin blockchain application is a public peer-to-peer payment application (see Appendix A for detailed information) that stores the transaction history on a digital database called a blockchain. Bitcoin is independent of an intermediary, such as a national bank (Ilk et al., forthcoming; Mattke et al., 2019). Bitcoin is an exclusively IS-enabled phenomenon because bitcoins are automatically generated by the Bitcoin blockchain application to reward the individual who first successfully executes the proof-of-work consensus mechanism (Nakamoto, 2008). This means that nobody can issue new bitcoins. Instead, new bitcoins

are an exclusively IS-enabled artefact generated through participation in the consensus mechanism or transaction process (see transaction process of bitcoin Mattke et al., 2018). Bitcoin is also an IS-governed phenomenon because transactions are validated and conducted without an intermediary and completely by the Bitcoin blockchain application (Mai et al., 2018). The use of and investment in bitcoin is decoupled from a central actor, such as a bank, and the Bitcoin blockchain application itself governs the use and ensures that exchange rules are enforced. For instance, in contrast to a fiat currency, where a bank is needed to prevent double-spending, the Bitcoin blockchain application itself regulates transactions and prevents double-spending, regulating and governing bitcoin without an intermediary (Nakamoto, 2008). Therefore, the use of bitcoin is always coupled with using the Bitcoin blockchain application. For instance, investing in bitcoin requires individuals to use the Bitcoin blockchain application to send and receive transactions. Investing in bitcoin is characterised as IS-governed because it is governed and regulated by an IS and where investing is coupled with using the same IS.

In summary, bitcoin is an IS phenomenon and investing in bitcoin is exclusively IS-enabled and IS-governed, so studying bitcoin investment from a IS research perspective is merited (Tarafdar & Davison, 2018).

2.2. Previous research into bitcoin

Extant research into bitcoin addresses technical topics such as self-organising in the Bitcoin community (Andersen & Bogusz, 2017) and anonymity in the Bitcoin blockchain application (Fabian et al., 2016; Yin et al., 2019). Further areas of research include behavioural bitcoin topics such as the use of bitcoin as means of payment (Abramova & Böhme, 2016), investment (Glaser et al., 2014) or resistance to using bitcoin as a means of payment (Mattke et al., 2018). Due to the behavioural nature of our research questions, our overview of research focuses on that stream.

Research into the use of bitcoin as a means of payment finds that most of the early adopters of the Bitcoin blockchain application were technology enthusiasts (Lindman et al., 2017). Early adopters were positively influenced by the expectation of faster transaction speed and the expectation of security and control of money (Abramova & Böhme, 2016). Another motivation for using bitcoin as a means of payment is that it is decentralised (Abramova & Böhme, 2016). Related research explaining why individuals resist using bitcoin as a means of payment identifies various motivations among different types of individuals and finds that the relative importance of the reasons for resisting varies among individuals (Mattke et al., 2018). The motivations to resist using bitcoin as a means of payment include, among others, transition costs, uncertainty costs, loss aversion, sunk costs, anticipated regret and decisional control. While findings show that anticipated regret is relevant for all individuals resisting using bitcoin as a means of payment, perceived uncertainty costs or transitions costs can also lead individuals to resist using bitcoin as a means of payment. Some individuals also perceive loss aversion, weighting losses associated with bitcoin use higher than the gains, which leads to resistance (Mattke et al., 2018). Beyond research discussing the use of bitcoin as a mean of payment, related research has studied what motivates individuals to invest in bitcoin (Glaser et al., 2014). While some research shows that future bitcoin value is associated with bullish forum posts in social media (Mai et al., 2018), researchers posit that the desire to make profit is the primary motivation of investing in bitcoin (Glaser et al., 2014). However, this is not supported by practical experience (Berger, 2017; Lindman et al., 2017).

In summary, we see that research into behavioural bitcoin topics has mostly explained why individuals resist using bitcoin or use bitcoin as a mean of payment. Less is known about what motivates individuals to invest in bitcoin, which would complement the research. The following section outlines the theoretical background of our study.

3. Theoretical background

This research identifies what motivations, in terms of perceptions about bitcoin, personal factors and regret sentiments, drive individual investment in bitcoin. We begin by presenting regret theory and describe how we apply its main tenets to theorise how motivations related to bitcoin influence bitcoin investment. Finally, we elucidate the research strand of investment.

3.1. Regret theory

Research shows that investment decisions are based on several conventions which capture the essence of rational behaviour under uncertainty (Kahneman & Tversky, 1979). Among the many different theories used in investment research, *regret theory* is particularly concerned with the uncertainty aspect (Loomes & Sugden, 1982). Regret theory extends other theories applied to study rational behaviour under uncertainty by considering additional relevant motivations called regret sentiments. Regret theory is useful in explaining observed axioms and violations of classical rational behaviour approaches (Michenaud & Solnik, 2008). Regret theory has been applied in various areas of research (for an overview see Appendix B), such as portfolio investment (Solnik & Zuo, 2012), IT investment (Lankton & Luft, 2008; EH Park et al., 2016),

demand for insurance (Braun & Muermann, 2004), allocation in pension schemes, hedging (Michenaud & Solnik, 2008), auctions (Engelbrecht-Wiggans & Katok, 2008) and individual purchase behaviour (Nasiry & Popescu, 2012).

The above studies are in line with the original regret theory developed by Loomes and Sugden (1982) positing that individual decision-making is driven by three basic types of motivation (see Table 1): perceptions about the investment object, personal factors, and regret sentiments, which indicates that individuals do not always decide rationally. These three motivations are valuable in understanding decision making in the context of investments.

3.2. Research on investments

Previous research focusing on financial investments rooted in financial research (Subrahmanyam, 2008) defines investment as a purchase of an asset in the hope that the asset will generate income or can be sold at a higher price in the future. We next explain how previous investment literature can be classified into the three types of motivations (grouped according to context) known from regret theory.

The expected monetary or non-monetary rewards of the investment object are usually the main drivers for investments (Beal et al., 2005; de Bondt, 1998; McLachlan & Gardner, 2004; Nagy & Obenberger, 1994). Various indicators of a sound financial return, such as dividends, yields, expected returns, financial stability, interest rates or price trends influence investment decisions (Baker & Haslem, 1974; Clark-Murphy & Soutar, 2005). Besides the expected rewards, research indicates that the values of the organisation (Aspara & Tikkanen, 2008) and ethical perceptions of the investment object (McLachlan & Gardner, 2004) influence investment decisions. However, even if individuals place value in ethical aspects, they only invest if their essential financial needs are satisfied through the investment (Mackenzie & Lewis, 1999). Even concerning socially responsible investments, individuals perceive the return on investment as equally important as non-socially responsible investments (McLachlan & Gardner, 2004). Research shows that despite valuing ethics and social responsibility, expected financial returns remain the main driver of investment decisions (Beal et al., 2005; Lewis & Mackenzie, 2000; Mackenzie & Lewis, 1999; McLachlan & Gardner, 2004) and that investments are not influenced by altruism but by reward motivation and self-interest (de Bondt & Thaler, 1995; Bretschneider & Leimeister, 2017). Research furthermore shows that perception of the management, the industry, the brand and the product are relevant (Aspara & Tikkanen, 2008; Bi et al., 2017; Bretschneider & Leimeister, 2017; Clark-Murphy & Soutar, 2005; Frieder & Subrahmanyam, 2005). For instance, individuals prefer to invest in products and projects that match their needs and what they like (Bretschneider & Leimeister, 2017). Investments are often influenced by the perception about the management team, including their skills, ideas and characteristics (Hoegen et al., 2018).

Table 1. Three types of motivation according to regret theory.

Motivations	Definition (based on Loomes and Sugden (1982))
Perception about the investment object	Qualities of the investment object which influence investment decisions.
Personal factors	Individual qualities which influence investment decisions.
Regret sentiments	Feelings of regret which influence investment decisions.

Previous studies show that personal factors such as risk attitudes also influence investing (Cohn et al., 1975; Ehm et al., 2014; Loomes & Sugden, 1982). Risk-averse investors prefer lower-risk investments with safe profit, while risk affine investors prefer higher-risk investments. We additionally know that individuals' perceptions about the product and the industry of the investment object influence their investment decisions. Research shows that individuals prefer to invest in their area of expertise (Clark-Murphy & Soutar, 2005) or in companies they work for (Benartzi, 2001). Furthermore, trust in and knowledge about the stock market influences whether individuals invest in stocks (Balloch et al., 2015; van Rooij et al., 2011). Research also found that personal factors are relevant to investments on crowdfunding platforms. Among others, research shows that on crowdfunding platforms where individuals can observe each other's actions, the need for a positive image influences investment decisions (Bretschneider & Leimeister, 2017).

Previous investment furthermore explains how sentiments (Lucey & Dowling, 2005) and especially regret sentiments (Barber & Odean, 1999; de Bondt & Thaler, 1995; Rau, 2015) influence investment decisions. Individuals' feelings of regret often lead to a disposition effect, meaning that individuals hold assets that are losing value for too long and sell assets that are gaining value too soon (Fogel & Berry, 2006; Jain et al., 2015). The feeling of regret is also studied in the field of IS investment. Research shows that the anticipation of possible regret may hinder an organisation's IT investments (Lankton & Luft, 2008; EH Park et al., 2016).

As illustrated in Table 2, which summarises the motivations shaping investment, investments are traditionally studied in the following three contexts sharing similarities with investment in bitcoin: stock and equity investment, crowdfunding and initial coin offering. We additionally see that all three motivations have only been studied in the stock and equity

Table 2. Previous research on motivations for investments.

Author	Perception about the investment object	Personal factors	Regret sentiments	Context
Aspara and Tikkanen (2008)	Attitude to a company, idea and product			Stock and equity investment
Baker and Haslem (1974)	Financial indicators and statements			
Balloch et al. (2015)		Stock market literacy, trust in the stock market		
Barber and Odean (1999)			Regret avoidance	
Beal et al. (2005)	Financial and non-financial returns, social impact			
de Bondt (1998)	Financial indicators and statements			
Clark-Murphy and Soutar (2005)	Financial indicators, perception about management	Personal expertise		
Cohn et al. (1975)		Personal risk attitude		
Ehm et al. (2014)		Risk attitude		
Fogel and Berry (2006)			Anticipated regret	
Frieder and Subrahmanyam (2005)	Brand perception, product perception			
Jain et al. (2015)			Regret aversion	
Lewis and Mackenzie (2000)	Financial returns, perceived ethics			
Mackenzie and Lewis (1999)	Financial returns, perceived ethics			
McLachlan and Gardner (2004)	Return on investment, ethical perceptions	Other-oriented, moral intensity		
Nagy and Obenberger (1994)	Financial indicators and statements, perceptions about a company's product or service, perceptions about a company reputation and ethics	Personal financial needs		
Rau (2015)			Regret avoidance	
van Rooij et al. (2011)		Financial literacy		
Bi et al. (2017)	Perception of product quality			Crowdfunding investment
Bretschneider and Leimeister (2017)	Expected reward, attitude towards the investment	Need for recognition, image motivation		
Hoegen et al. (2018)	Monetary and non-monetary reward, perception about the founder	Investor characteristics		
Mollick (2014)	Project quality			
Vismara (2016)	Perception of the founder			
Adhami et al. (2018)	Perception about service, profit, quality of a project			Initial coin offering investment
Fisch (2019)	Technical indicators			
Fisch et al. (2019)	Ideological motives, technical motives, financial motives			

investment context. In the following section, we show how investment in bitcoin differs from previously studied investments and why traditional explanations for investments do not sufficiently explain the motivations for bitcoin investment (Mai et al., 2018).

3.3. Bitcoin investment with unique characteristics

While previous investment studies consider the context of stock and equity investment, crowdfunding and initial coin offering investments (see Table 3 for an overview and Appendix C for further explanation), investing in bitcoin differs from these investment contexts due to the unique characteristics of bitcoin.

Research shows that the bitcoin price is autonomous and does not depend on the price of traditional financial assets, such as stocks (Chen, 2018). Further, the bitcoin price is more volatile than fiat currencies, such as the US dollar, which limits the use of bitcoin as a store value (Böhme et al., 2015). This limits the applicability of research into fiat currencies to bitcoin. Furthermore, investment in fiat currencies is to a large extent influenced by the interest rates of the currency, geopolitical factors and the decisions of national governments (Melvin & Taylor, 2009). This does not hold for bitcoin, whose value is influenced by individuals. The low entry barriers for investing in bitcoin also attracts non-professional investors (Stray, 2017).

To guide investors, most previously studied investments have at least some validated and relevant investment information, such as ratings or credit scores by third parties and intermediaries (Hoegen et al., 2018), which are scarce for bitcoin. Bitcoin does not have an intermediary, such as a bank, because investing in bitcoin is an exclusively IS-enabled and IS-governed form of investment. The Bitcoin blockchain application thus appeals to individuals who support an ideology of decentralisation, the empowerment of the individual, and a weaker role of intermediaries, such as banks. This points to different motivations for investment behaviour, exposes individuals to new perceptions and confronts them with new issues. For instance, if a bitcoin private key is lost or stolen, there is no intermediary to help reset access to the bitcoins.

In bitcoin investment, individuals do not assess or rely on the competence of a management team or an organisation, but rather a decentralised blockchain application. The investment is not backed up by a company that generates revenue,

Table 3. Comparison of investing in bitcoin to previously studied forms of investments.

Investment Characteristic	Stock and equity investment	Crowdfunding investment	Initial coin offering investment	Bitcoin investment
Depicts an exclusively IS-enabled investment			✓	✓
Depicts an IS-governed investment			✓	✓
Is associated with persons in charge	✓(e.g., management)	✓(e.g., crowdfunding founder)	✓(e.g., founder of ICO)	
Aims to fund a venture	✓	✓	✓	
Has an intermediary	✓	✓		
Has an expected future cash flow	✓(e.g., dividends)	✓(e.g., discounted cash flow in equity crowdfunding or non-monetary rewards in reward-based crowdfunding)	✓(e.g., expected share of the profit through equity token or access to non-monetary rewards such as services through utility token)	
Acquires a stake	✓(e.g., investment in stocks)	✓(e.g., equity crowdfunding)	✓(e.g., equity tokens)	

nor does it have any tangible real-world value. Finally, investing in bitcoin is unique, as the proof-of-work consensus algorithm manages the supply of new bitcoins at a *"known, geometrically decaying rate, so demand from businesses and individuals represents the main driver"* (Mai et al., 2018, p. 24).

In summary, although the bitcoin investment context is similar to other investment contexts in some ways, its unique qualities make it different in significant ways. Due to these differences, we cannot apply the findings from extant related research into investment in other contexts directly to the context of bitcoin. The following section identifies the factors relevant and specific to bitcoin investments that motivate our mixed methods approach.

4. Mixed methods approach

We follow a mixed methods approach with two studies. We theorise that investment in bitcoin is shaped by three motivations (Loomes & Sugden, 1982): perceptions about bitcoin, personal factors and regret sentiments. Since the bitcoin investment context differs from the context of stock and equity investments, crowdfunding and initial coin offering, our objective is to identify configurations of motivations that drive bitcoin investment based on perceptions about bitcoin, bitcoin-specific personal factors and regret sentiments relevant to bitcoin investment.

Study 1 takes a qualitative approach to answer RQ1 by enriching the secondary data about motivations gained from the literature with primary data on bitcoin-specific factors. We know from previous studies that individual behaviour is complex and not determined by a single motivation. Rather, individual behaviour is grounded in the presence and/or absence of single motivations in an asymmetric relationship to each other constituting configurations that drive individual behaviour (Liu et al., 2017; Ordanini et al., 2014). The relative importance of motivational factors varies from individual to individual, so individual behaviour can be grounded in different configurations of motivations, reflecting equifinality (Liu et al., 2017). Extant investment research has already shown that investors differ in terms of what motivates their investments decisions (Clark-Murphy & Soutar, 2005). Bitcoin research also confirms that individual investment behaviour is also shaped by multiple configurations of motivations (Mattke et al., 2018). Beyond identifying specific motivations, it is necessary to identify sufficient configurations of these motivations that collectively drive bitcoin investment (see Figure 1). To achieve this, Study 2 builds upon the results of Study 1 and takes a configurational approach to identify distinct configurations of the motivations identified in

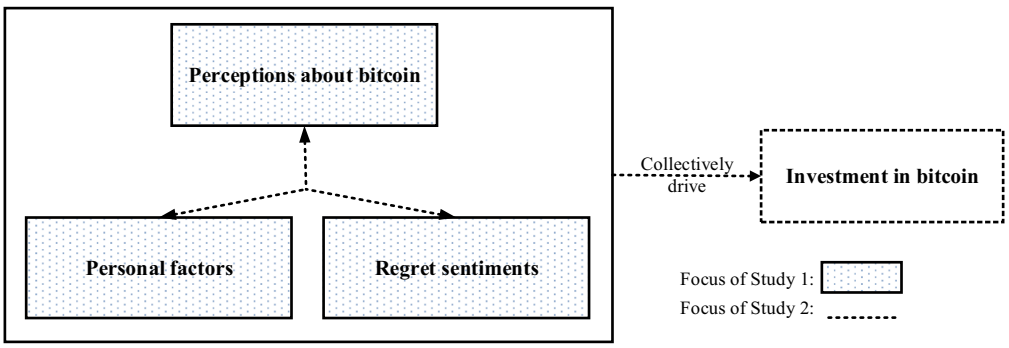


Figure 1. Research approach and focus of Study 1 and Study 2.

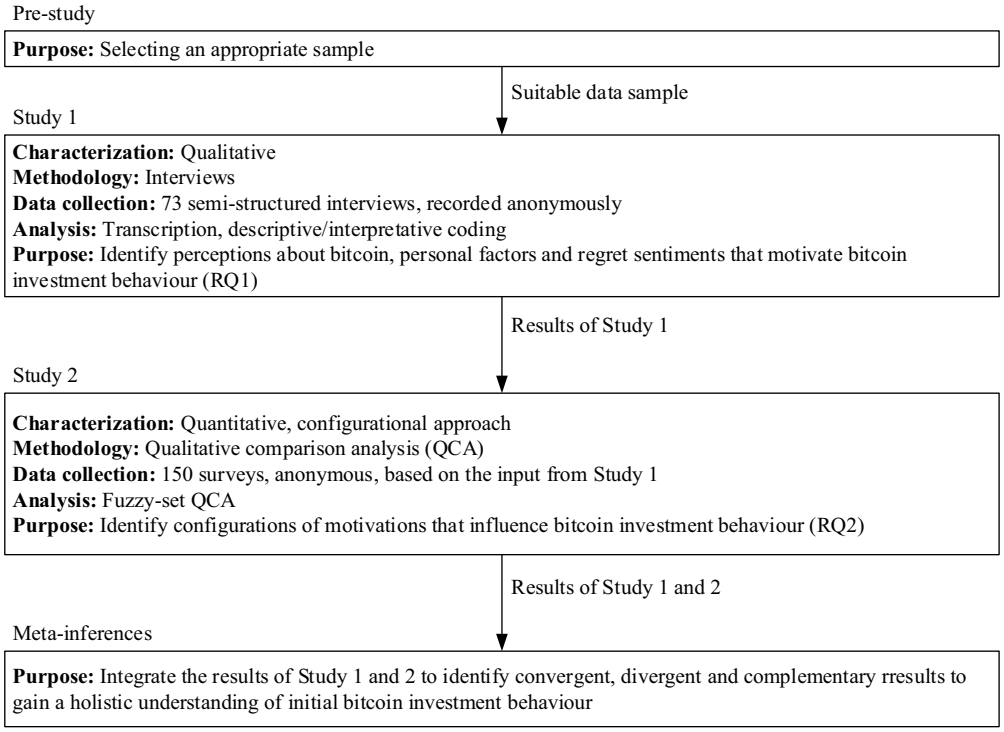


Figure 2. Mixed methods approach.

Study 1 which drive bitcoin investment, thus answering RQ2.

To integrate the results gained from Study 1 and Study 2, we need to overcome the framing of pure qualitative or quantitative studies and therefore frame our research with a mixed methods approach (Venkatesh et al., 2013). Following the lead of other mixed methods research (Müller et al., 2017), we adopt a sequential, two-strand, equally-weighted research design (see Figure 2) following a developmental purpose to investigate the motivations that collectively drive the investment in bitcoin. We additionally conduct a pre-study to select the appropriate sampling strategy for both studies. We adopt the single worldview framed by regret theory (Loomes & Sugden, 1982) and apply the single epistemological perspective of pragmatism to select the most suitable research approach to answer each research question. We then triangulate the results from Study 1 and Study 2 to deduct convergent, divergent and complementary insights to better understand the motivations driving bitcoin investment.

4.1. Pre-study

To develop a sampling strategy, we undertook a pre-study with six participants who had never invested in bitcoin and with six participants who had invested in bitcoin. By interviewing participants who had never invested in bitcoin (pre-adoption phase) (Maier, 2020) we gained insights into why individuals would invest in bitcoin for the first time. By interviewing participants who had invested in bitcoin and still owned bitcoin (in the use phase), we gained insights into why individuals would invest in bitcoin again based on previous bitcoin investment experience. For instance, a participant reported her initial assessment that she had the necessary skills to invest in bitcoin and to earn profits. After having invested in bitcoin, her motivation for once again investing in bitcoin changed, and satisfaction and confirmation of her initial perceptions were the main drivers for repeated bitcoin investments. This illustrates that gathering data from participants with bitcoin investment experience is suited for explaining their reinvestment, but not for explaining why individuals invested in the first place. Since we want to understand what perceptions about bitcoin, personal factors and sentiments, grouped in configurations, motivate first-time bitcoin investment, we concluded for the sampling strategy to concentrate in both studies on participants who have never invested in bitcoin before but are aware of bitcoin.

4.2. Study 1: motivations driving investment in bitcoin

In Study 1 we conducted 73 interviews to collect qualitative data on specific motivations, in terms of perceptions about bitcoin, bitcoin-related personal factors and regret sentiment, driving bitcoin investment (RQ1). This section outlines our data collection procedure, the data analysis process, the validation and the results.

Table 4. Participant demographics (N = 73).

Gender in per cent	Male	57.53
	Female	42.47
Age in per cent (mean = 34.27; standard deviation = 12.48)	18–20	16.44
	21–30	54.79
	31–40	12.35
	41–50	8.21
	>50	8.21

4.2.1. Data collection

We conducted 73 semi-structured interviews (see demographic information in Table 4). In line with our sampling strategy, we pre-screened potential participants and only interviewed individuals who had never owned cryptocurrency but were aware of bitcoin. The interviews lasted about 15 minutes each and were conducted mainly at blockchain and bitcoin information events in January 2018. We gained further participants by applying a snowball system. Our interview guideline (see Appendix D) was based on regret theory. At the beginning of the interview, we asked an open question about why the participants might invest in bitcoin and about the circumstances influencing their decision. We then asked participants what motivations are relevant to their decision to invest in bitcoin, asking follow-up questions prompting participants to complete or clarify their answers and provide more details on what would influence their decision. With the agreement of the participants, we recorded the interviews, collecting approximately 234 hours of audio data. Afterwards, we anonymised and transcribed the interviews, resulting in approximately 200 pages of transcription.

Table 5. Overview of the four-step analysis approach.

Step	Procedure
Step 1. Descriptive coding	Identify passages and assign descriptive codes
Step 2. Categorising into the three motivations	Categorise descriptive codes into "perceptions about bitcoin", "personal factors" and "regret sentiments"
Step 3. Interpretive coding	Apply interpretive coding to the descriptive codes.
Step 4. Formal review of the interpretive coding	Ensure consistent coding and compute inter-judge reliability.

Table 6. Validation of qualitative inferences.

Category of validity	Validation
Design validity	• Precise description of research process ensures descriptive validity • Sufficiently large sample of relevant target population (Sarker et al., 2013) ensures credibility and transparency • Applying well-established theory ensures transferability • Excluding individuals who have invested ensures focus on pre-adoption phase of bitcoin investment (Maier, 2020)
Analytical validity	• Interview structure based on regret theory (Loomes & Sugden, 1982) ensures plausibility and theoretical validity • Using semi-structured approach leaves room for unforeseen questions but preserves consistency within the structure • Pretesting interview questions with students ensures clarity • Exact and complete transcripts of interviews ensures completeness
Inferential validity	• Mirroring answers back to participants ensures correct understanding of descriptions and feelings and greater interpretative validity • Coding drawn directly from the answers by applying descriptive and interpretive coding (Myers, 2013) • Coding validated by multiple judges (Randolph, 2005)

4.2.2. Data analysis

We took a four-step analysis approach (see Table 5) to code the transcribed interviews using MAXQDA 12 and reveal perceptions about bitcoin, personal factors and regret sentiments that influence bitcoin investment. See Appendix E for coding examples and further details of the four-step analysis approach.

4.2.3. Validation

We validated the qualitative inferences in terms of data collection and analysis within different categories (see Table 6).

4.2.4. Results

The purpose of Study 1 is to identify perceptions about bitcoin, personal factors and regret sentiments, which are components of configurations of motivations that drive investment in bitcoin (RQ1). Our findings are displayed in Figure 3.

As the result of the data analysis, we revealed three overall perceptions about bitcoin investment

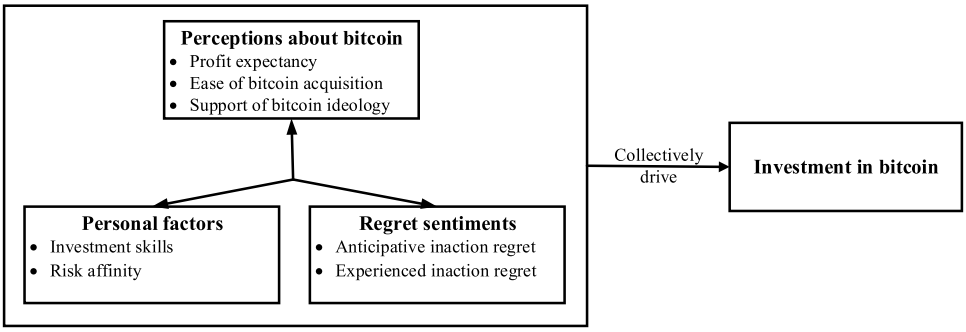


Figure 3. Findings clustered based on regret theory.

Table 7. Definition of the identified motivations relevant for investing in bitcoin.

	Motivation	Contextual definition (based on; context)	Exemplary quotations
Perception about bitcoin	Profit expectancy	The degree to which an individual expects to benefit financially from investing in bitcoin (Ali, 2011; East, 1993; context: investment).	"The chance to make profit . . . meaning you can easily make so much profit, that's what motivates me." "The main reason to invest in bitcoin is clearly that I expect that bitcoin's value will increase in the next years."
	Ease of bitcoin acquisition	The degree to which an individual expects the process of acquiring bitcoin to be easy (based on Pavlou and Fygenson (2006); context: electronic commerce).	"Investing in bitcoin is easy and uncomplicated." "I can invest even a small amount, that's what makes the investment attractive."
	Support of bitcoin ideology	The degree to which an individual supports the ideology of bitcoin (Karahanna et al., 2006; context IS adoption)	"Investing in bitcoin is like being part of modern history. I feel this is the next big step, to cut out the middleman and form a global currency." "I believe in the ideology of Bitcoin and I trust the concept of Blockchain, which is why I would invest in Bitcoin."
Personal factors	Investment skills	The degree to which an individual has knowledge about investments and how to make sound investments (Pavlou & Fygenson, 2006; context: electronic commerce).	"I have a good feeling for investments and I mostly know where to invest. So, I will invest in bitcoin." "In the last years, it has proven that I got a good sense to do the right investments, so that motivates me to invest."
	Risk affinity	The degree to which an individual as the tendency to make risky investments (Allen et al., 2005; Slovic, 1972; context: risk taking behaviour).	"I am willing to take the risk on investments, so that's definitely why I invest. I think taking risk is normal for such an investment." "I am the type of guy who sometimes takes a risk and that is the same with the decision to invest in bitcoin. I see the risk as normal for such an investment."
Regret sentiments	Experienced inaction regret	The degree to which an individual has the feeling of regret not having invested in bitcoin in the past (Zou et al., 2015; context: post adoption).	"I was thinking about investing in bitcoin in early 2011 and did not take the chance. That's why I want to invest now." "I was confronted with the decision whether to invest in Bitcoin in early 2015 and I decided not to invest. I could have been millionaire today, which is what I regret. At least I will invest in bitcoin now."
	Anticipated inaction regret	The degree to which an individual anticipates regretting in the future not having invested in bitcoin (Nasiry & Popescu, 2012; Tsiros, 2008; context: purchase behaviour).	"I do not want to miss the chance now, or I will regret the decision in the future." "I truly believe that I would one day regret my decision to not invest in Bitcoin, and I don't want to regret that. So, I think I will invest some of my money in Bitcoin."

(profit expectancy, ease of bitcoin acquisition, support of bitcoin ideology), two personal factors (investment skills, risk affinity) and two regret sentiments (anticipated inaction regret, experienced inaction regret) which drive investment in bitcoin. In Table 7 we explain each motivation that influences investment in bitcoin, we provide a definition and provide two exemplary quotations from the interviews. See Appendix E for a complete list of the descriptive codes and additional quotations from the interviews.

4.3. Study 2: configurations resulting in investment in bitcoin

As indicated in Figure 2, the seven motivations identified in Study 1 serve as a basis for Study 2.

4.3.1. Data collection

For the data collection, we prepared an online survey and advertised for participants on Amazon Mechanical Turk (mTurk). Using mTurk for data collection is a well-established approach in IS research, is considered as effective as traditional data collection approaches (Maier et al., 2019). Recent research shows that acquiring participants from mTurk does not create demand effects (Mummolo & Peterson, 2018). In Study 2, we filtered participants by country in mTurk to ensure that no one who participated in Study 1 also participated in Study 2.

Our sampling strategy was aligned with our pre-study (see section 4.1) and with the sampling strategy of Study 1. To implement this sample strategy, we used multiple pre-screening questions (e.g., "Do you currently own bitcoin?", "Have you bought bitcoin before?") to filter participants.

Overall, 528 participants took part in the survey, 179 of whom were eliminated because they currently own bitcoin and 124 more were eliminated because they were not aware of bitcoin. A further 18 participants did not answer the survey completely and were consequently removed from the sample and 12 participants were removed because they failed at least one attention-trap question. The average time for completing the survey was eight minutes with a standard deviation of four minutes. Five participants were removed from the sample because they completed the survey in under 100 seconds, which is unrealistically quick. The final sample consists of 150 participants, whose demographics are

Table 8. Participant demographics (n = 150).

Highest education level in per cent	Less than High School	0.67
	High School/GED	4.67
	Some College	14.00
	2-year College Degree	9.33
	4-year College Degree	38.00
	Master's Degree	30.00
	Doctoral Degree	2.00
	Professional Degree (JD, MD)	1.33
Gender in per cent	Male	55.33
	Female	44.67
Age in per cent (mean = 28.95; standard deviation = 5.99)	18–20	0.00
	21–30	57.33
	31–40	22.00
	41–50	14.67
	>50	6.00
Income (US$) in percent (mean 21,426)	<12,000	41.67
	12,000–19.999	7.41
	20,000–29,999	16.67
	30,000–39.999	13.89
	40,000–49,999	6.48
	> 50,000	13.88

summarised in Table 8. We also tested for common method bias (see Appendix F), and determined that common method bias is not an issue in this study.

4.3.2. Measures

Our survey measures were based on measures used and validated in previous research or adapted from measures from related research. For *intention to invest in bitcoin*, we drew on two items from investment literature (East, 1993). For *profit expectancy*, we also drew on five items from investment literature (Ali, 2011). For *ease of bitcoin acquisition*, we adapted two items from previous research about individuals purchase behaviour in electronic commerce (Pavlou & Fygenson, 2006), changing the product focus of the items to a focus on the acquisition of bitcoin. For *investment skills*, we adapted three items from studies of individual online purchasing behaviour in electronic commerce (Pavlou & Fygenson, 2006) by adapting the skill of making good online purchases to the skill of making good investments. For *support of bitcoin ideology*, we adapted four related items from technology acceptance research (Karahanna et al., 2006) to the context of bitcoin by changing the focus from CRM systems and computer to bitcoin and blockchain applications. Based on research in psychology (Allen et al., 2005), we used five items for *risk affinity*. For *experienced inaction regret*, we adapted two items focusing on experienced regret developed by Inman and Zeelenberg (2002), by adapting from a general decision to the decision to invest in bitcoin. For *anticipated inaction regret*, we used a further item developed by Inman and Zeelenberg (2002) and items developed by Tsiros (2008), which focus on anticipated regret in purchase decisions. All items are displayed in Appendix G.

4.3.3. Measurement model

To use the empirical data for data analysis, we first need to test the measurement model for indicator reliability, construct reliability and discriminant validity (Bagozzi, 1979), which is also common for QCA studies (Liu et al., 2017). For each item, the loading is at least 0.707, which means that each item explains at least 50 per cent of the variance of the latent variable and indicates *indicator reliability* (Carmines & Zeller, 2008). As illustrated in Appendix G, *construct reliability* is ensured as the average variance extracted (AVE) is higher than 0.50 and composite reliability (CR) is higher than 0.70 (Fornell & Larcker, 1981). We determined *discriminant validity* by examining the square root of the AVE, which fulfilled the condition of being greater than the corresponding construct correlations (Fornell & Larcker, 1981; Hulland, 1999) (see Appendix G) We additionally tested the heterotrait-monotrait (HTMT) ratio, calculating the highest value of 0.81, which is lower than the absolute $HTMT_{0.85}$ criterion (Henseler et al., 2014). That result proves the discriminant validity of the measurement model. These tests illustrate that our measurement model is valid and can be used for further analysis.

4.3.4. Data analysis using fsQCA

We took a configurational approach (Misangyi et al., 2017; Park et al., forthcoming) and used fuzzy-set qualitative comparison analysis (fsQCA) to evaluate the data and to answer the second research question, which asks about configurations of perceptions about bitcoin, personal factors and regret-sentiments resulting in investment in bitcoin (RQ2).

fsQCA is based on set theory and enables us to analyse the relationship between configurations of conditions and the outcome. In this study, the conditions consist of the three perceptions about bitcoin, the two personal factors and the two regret sentiments. The outcome is investment in bitcoin. Consequently, fsQCA allows us to study causal complexity (Misangyi et al., 2017) as we can examine the asymmetric relationship between conditions and equifinality. Asymmetric relationship between conditions means that investment in bitcoin can be grounded in the presence and the absence of single conditions, depending on the other conditions in a configuration. Furthermore, some conditions can be overridden in a configuration, so that those conditions do not relate to bitcoin investment, which implies that it is irrelevant if the condition is present or absent. Thus, multiple conditions, which may be present or absent, grouped into configurations drive investment in bitcoin. Equifinality means that individuals perform the same behaviour for distinct

reasons (Misangyi et al., 2017). In this case, individuals starting with different perceptions about bitcoin, personal factors and regret sentiments and following many different configurations may all ultimately invest in bitcoin. Thus, fsQCA enables us to study instantiations of a general model and thereby consider the heterogeneity of the individuals.

4.3.5. Data analysis

We used the fsQCA software (Ragin, 2006b) to analyse our data. First we calibrated the interval scale values from the survey into fuzzy-sets following recommendations in QCA literature and applying a well-established calibration process (Misangyi et al., 2017). In the calibration, we set the anchor for full non-membership at the value 1, the anchor for the crossover point at the value 4 and the anchor for full membership at the value 7. To avoid fuzzy-set memberships of 0.50, we followed previous research and added a small constant of 0.001 (Fiss, 2011). To ensure that this did not disturb the results, we reran the analysis but subtracted the small constant of 0.001 (Crilly et al., 2012). The results stayed the same, which showed that the results were not distorted by adding the small constant to avoid 0.50 fuzzy-set membership.

Using a fsQCA, we tested for *necessary conditions* first. A necessary condition is a condition that is always exists when the outcome exists. In this study, we used the recommended consistency threshold of 0.90 (Schneider & Wagemann, 2012). If a single condition has a greater consistency value than the consistency threshold of 0.90, it depicts a necessary condition. The consistency thereby indicates the degree to which configurations with the same conditions share the same outcome (Ragin, 2006a). To avoid trivial necessary conditions (type 1 error) we additionally tested whether the coverage score and the relevance of necessity (RoN) score are higher than 0.60 (Ragin, 2006a). Both tests show scores higher than 0.60. We then tested for *sufficient configurations*. If a sufficient configuration of conditions is present, it brings about the outcome (Ragin, 2014). In other words, whenever a sufficient configuration exists, the outcome will exist as well.

To conduct the test for sufficient configurations, we took a four-step approach. First, using the calibrated data, we prepared a truth table data structure. In a truth table data structure, each condition and outcome are displayed in one column and each row contains one observation. This means that a truth table data structure lists all configurations found in the data set. The truth table data structure indicated that we have a nearly equal number of configurations with presence and absence of intention to invest in bitcoin, thus we can say that the results were not distorted by an unbalanced data set. Second, we consolidated the truth table data structure by summarising similar configurations together and calculating the consistency. The configurations account for more than 40 per cent of the possible configurations, which shows that the data set has high explanatory power. Third, we applied consistency thresholds and a frequency threshold, which reduced the truth table to sufficient configurations. In line with recommendations from literature, we used consistency thresholds of 0.80, which is higher than the minimum threshold of 0.75 (Ragin, 2006a) and frequently used in QCA studies (e.g., Park & Mithas, 2020). We used a frequency threshold of five because a higher frequency threshold is recommended for large sample size (Maggetti & Levi-Faur, 2013) and the frequency threshold of five is often used in QCA studies (e.g., see Lee et al., 2019). This means that only configurations with consistency levels greater than consistency thresholds and configurations that exceed the frequency threshold are considered sufficient configurations. Fourth, we applied the Quine-McCluskey algorithm to simplify the sufficient configurations using Boolean logic (Ragin, 2014). This resulted in a solution formula, which described the underlying pattern of the truth table.

4.3.6. Validation

We evaluated the quantitative inferences from the configurational approach in terms of data collection and analysis within different categories of validity (see Table 9).

4.3.7. Results

The analysis of necessary conditions reveals investment skills as a necessary condition (consistency > 0.93, coverage > 0.82, RoN >0.70) for bitcoin investment (see Appendix H for further details). This means that individuals need to have knowledge about

Table 9. Validation of quantitative inferences.

Category of validity	Validation
Design validity	• Constructs were adapted from existing literature (Appendix G) • Sufficient sample size and representative population (Marx, 2010) • Based on regret theory (Loomes & Sugden, 1982) to ensure internal validity and highly generalisable design to ensure external validity • No common method bias in this study (Appendix F)
Measurement validity	• No issues with content validity, indicator reliability, construct reliability and discriminant validity (Bagozzi, 1979) • Robust to a shift of calibration anchors (Ordanini et al., 2014; Park et al., 2017) • Robust to adding a small constant to fuzzy-set membership of 0.50 (Fiss, 2011)
Inferential validity	• We measured statistical conclusion validity with appropriate tools common in QCA research (Ragin, 2006b) • Reliable and robust solution through the use of high consistency and frequency thresholds (Schneider & Wagemann, 2012)

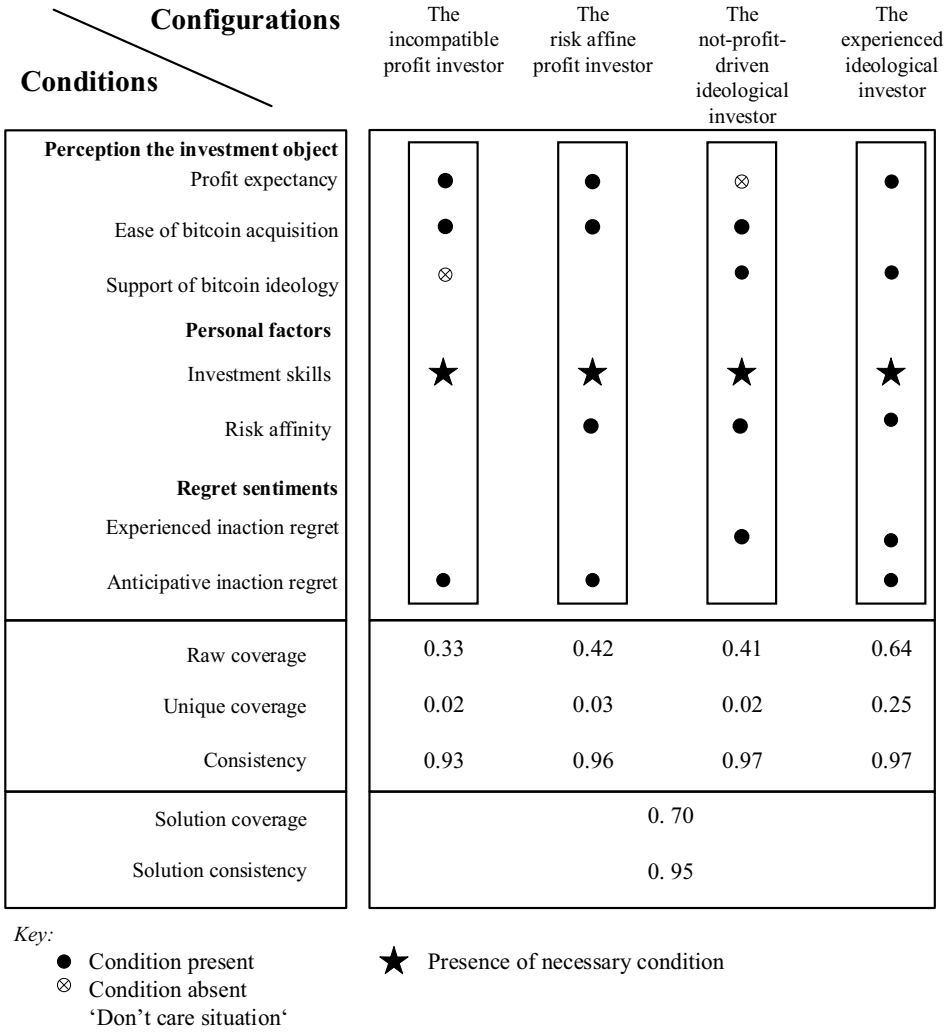


Figure 4. Four bitcoin investment configurations.

Table 10. Consistency and coverage scores of the QCA solution.

Score	Explanation of the scores (Ragin, 2006a; Schneider & Wagemann, 2012)
Raw coverage	Quantifies the proportion of configurations in the data set explained by a single sufficient configuration.
Unique coverage	Quantifies the proportion of configurations uniquely explained by a single sufficient configuration excluding the proportion explained by other sufficient configurations.
Consistency	Quantifies the extent to which a single sufficient configuration results in investing in bitcoin.
Solution coverage	Quantifies the extent to which the four sufficient configurations together result in bitcoin investment.
Solution consistency	Quantifies the extent to which the four sufficient configurations explain the configurations in the data set.

investments and how to make sound investments to invest in bitcoin. This does not imply that investment skills always lead to investment, which is addressed by an analysis of sufficient configurations. The analysis of sufficient configurations reveals a Boolean expression, which is graphically displayed in Figure 4 and the explanations of the different consistency and coverage scores are summarised in Table 10.

The four sufficient configurations have a solution coverage of 0.70, which indicates a high empirical relevance of the four sufficient configurations, and a solution consistency of 0.95, which indicates a high consistency. Figure 4 below graphically represents the sufficient configurations. A black circle indicates the presence of a condition, a white crossed-out circle indicates the absence of a condition. A blank space indicates a "don't care" situation, in which the respective condition may be present or absent, meaning that the condition is overridden by one or more other conditions and does not influence individual investing in bitcoin. Additionally, a black star indicates the presence of a necessary condition. Overall, we identify four distinct sufficient configurations that drive investing in bitcoin.

The first configuration depicts an investor who expects to make profit by investing in bitcoin and appreciates the advantage of the easy bitcoin acquisition process. This type of investor does not

support the values embodied by bitcoin. Furthermore, the investor anticipates inaction regret if she does not invest in bitcoin. For this type of investor, experienced inaction regret does not influence the decision to invest in bitcoin and it does not matter whether the investor is risk affine. We, therefore, label this type of investor *the incompatible profit investor*.

The second configuration presents a risk affine investor that sees the advantages of bitcoin in terms of expected profit and in the ease of the acquisition process. For this type of investor, the ideology embodied by bitcoin does not influence the investment in bitcoin. Anticipated inaction regret influences the investment of this investor. We therefore label this type of investor *the risk affine profit investor*.

The third configuration is a risk affine investor who does not expect to make profit through the investment in bitcoin, but who supports the bitcoin ideology. This type of investor assesses the acquisition process as easy and has strong investment skills. In contrast to the other investor types, this investor is driven by experienced inaction regret, meaning that she regrets former decisions not to invest in bitcoin. Anticipated inaction regret does not influence the investment in bitcoin of this investor. We therefore label this investor *the not-profit-driven ideological investor*.

The fourth configuration depicts a type of investor that expects profit from the investment and supports the bitcoin ideology. In contrast to the other investor types, the ease of bitcoin acquisition does not influence the investment decision. This investor shows risk affinity and investment skills and is the only investor driven by experienced and anticipated inaction regret. We hereafter label this type of investor *the experienced ideological investor*.

All four sufficient configurations have a consistency score which is higher than the minimum required consistency of 0.75. The raw coverage ranges from 0.33 to 0.64, indicating that all four sufficient

Table 11. Triangulation of qualitative and quantitative inferences to meta-inferences.

Qualitative inferences of Study 1 (see Table 7)	Quantitative inferences of Study 2 (see Figure 4)	Meta-inferences (Study 1 and Study 2)	Explanation (Note: No explanation needed when results of Study 1 and Study 2 are convergent; indicated by a crossed-out field)
Perceptions about bitcoin			
Profit expectancy is a motivation for investment in bitcoin.	Complementary to Study 1. One sufficient configuration leading to bitcoin investment lacks profit expectancy.	Not all individuals who invest in bitcoin are motivated by profit expectation.	Bitcoin relates to an ideology of the empowerment of individuals and the independence of intermediaries (Nakamoto, 2008) and the ideology behind bitcoin is a driver for individual behaviour (Lindman et al., 2017). This contrasts with prior research on investment, where profit expectation is even necessary for ethical investments (Mackenzie & Lewis, 1999).
Ease of bitcoin acquisition is a motivation for investment in bitcoin.	Convergent with Study 1. Showing that ease of bitcoin acquisition is present in three out of four sufficient configurations.	Most individuals investing in bitcoin value the easy acquisition of bitcoin.	
Support of bitcoin ideology is a motivation for investment in bitcoin.	Complementary to Study 1. Support of bitcoin ideology is present in two out of four, absent in one and does not influence one sufficient configuration.	Not all individuals investing in bitcoin are influenced by the motivation to support the bitcoin ideology.	Prosocial motives, such as support of bitcoin ideology, and self-interested motives, such as profit expectancy, are not mutually exclusive but can also coexist (Bretschneider & Leimeister, 2017).
Personal factors			
Investment skills are a motivation for investment in bitcoin.	Complementary to Study 1. Study 2 additionally reveals investment skills as a necessary condition.	Only individuals with investment skills will invest in bitcoin.	The importance of investment skills can be explained by the relatively new form of investment and its high volatility (Mai et al., 2018).
Risk affinity is a motivation for investment in bitcoin.	Convergent with Study 1. Showing that risk affinity is relevant in three out of four sufficient configurations.	Most investor types require risk affinity.	
Regret sentiments			
Experienced inaction regret is a motivation for investment in bitcoin.	Complementary to Study 1. Relevant in two out of four sufficient configurations, where support of bitcoin ideology is present as well.	Individuals who invest in bitcoin and support the bitcoin ideology also perceive experienced inaction regret.	The influence of experienced regret is coupled with the support of the bitcoin ideology. This suggests that only individuals who support the ideology of bitcoin feel regret that they did not invest earlier and thus did not support the ideology of bitcoin earlier.
Anticipated inaction regret is a motivation for individuals.	Complementary to qualitative inferences. Also indicating a contribution of anticipated inaction regret in three out of four sufficient configurations, where profit expectancy is present as well.	Individual investing in bitcoin and expect profit also perceive anticipated inaction regret.	Anticipated inaction regret occurs when individuals think prospectively (EH Park et al., 2016). The expected return of bitcoin is anchored in the future and individuals who expect a return are motivated by regret if they do not invest.

configurations have high empirical relevance. The unique coverage is higher than 0.00 and ranges from 0.02 to 0.25, which indicates that every sufficient configuration uniquely contributes to explaining investing in bitcoin. The difference of the unique coverage and raw coverage indicates that *the experienced ideological investor* is the most distinct configuration, meaning that the explanation of this configuration overlaps only slightly with the other configurations (Wagemann et al., 2016).

4.4. Meta-inferences

In this section, we present the meta-inferences deduced from the triangulation of qualitative inferences from Study 1 and quantitative inferences from Study 2 and their validity.

4.4.1. Deduction of meta-inferences

Throughout this manuscript, we adopted the epistemological perspective of pragmatism, as it places the highest importance on selecting suitable methods to answer the research questions and enables us to select combined reasoning approaches to answer both research questions without a purely qualitative or quantitative frame (Venkatesh et al., 2013). We used inductive reasoning to identify relevant motivations which are components of configurations that drive investing in bitcoin and abductive reasoning (Park et al., forthcoming) to identify configurations of those motivations driving investing in bitcoin. The mixture of both modes of generalisability enables us to triangulate both inferences and deduce meta-inferences (see Table 11).

The qualitative and quantitative inferences were largely convergent, which shows a good quality of inferences and strengthens the results (Venkatesh et al., 2013). We also deduced complementary results, such as the presence of investment skills not only being important, as expected from Study 1 but also a necessary driver of bitcoin investment. Where the inferences reveal complementary or even divergent results, we sought theoretically sound explanations to assess the credibility of meta-inferences (see Table 11). Where the results are convergent, no additional explanation is needed.

With our mixed methods approach, we see that Study 1 could only provide a general explanation of motivations driving investment in bitcoin. With Study 2, which is based on the motivations identified in Study 1, we see that the motivations are interconnected and that investment in bitcoin can only be explained for instantiations of the deduced model. Thus, a different instantiation of the deducted model explains the investment of different individuals. The richness and robustness of results prove the credibility of the identified meta-inferences.

4.4.2. Quality of meta-inferences

This section describes our assessment of the design and explanation quality of our meta-inferences (Venkatesh et al., 2013), as summarised in Table 12 below.

5. Discussion

Millions of individuals have invested in bitcoin over the past years (Lielacher, 2019), but certain qualities of

Table 12. Quality of meta-inferences.

Category of validation	Validation
Design quality	• Mixed methods strategy encompassing suitable methods, here interviews and QCA, to answer the research questions. • Interviews and QCA was applied with quality and rigour, as stated in the validity sections (Myers, 2013; Schneider & Wagemann, 2012).
Explanation quality	• Triangulation of both studies revealed reliable theoretical statements of when and how motivations yield investment in bitcoin. • Meta-inferences are transferable and aim at a holistic view, as we combined our inductive and abductive reasoning approaches and revealed general theoretical statements of bitcoin investment. • High convergence of results showing strength and quality of inferences and even complementary results completing the holistic understanding of bitcoin investment (see Table 11).
Legitimation of meta-inferences and potential threats and remedies	We identified potential threats to and illustrated the legitimacy of our meta-inferences with suitable remedies in several categories: • sample integration: we undertook sequential collection of qualitative and quantitative data within the same population, but with disjunctive samples. • *inside-outside*: different members of the research team independently validated our coding and we calculated interrater reliability. • *weakness minimisation*: we complimented the subjective results from the interviews with QCA to enhance objectivity. • *conversion*: the conversion of our qualitative inferences was based on the framing theory of regret theory and double-checked with the participants. • *multiple validity*: we assured the validity of the interview study, the QCA study and the mixed-methods approach individually and separately. • *political validity*: the meta-inferences were drawn from the triangulation of the interview study and the QCA study. The results supported and enlarged regret theory and led to answers to our research questions motivations and configurations of motivations driving bitcoin investment.

the phenomenon differentiate it the context of stock and equity investment, crowdfunding or initial coin offering investments (see Table 2). By characterising investment in bitcoin as an exclusively IS-enabled and IS-governed phenomenon, we anchor our research as core to IS research. We follow a mixed methods approach with two sequential studies and draw on regret theory. In Study 1, we collect qualitative data using interviews to identify seven motivations which are components of configurations of motivations that drive bitcoin investment, to address RQ1. Based on those findings, we collect quantitative data to reveal four configurations of these motivations that drive bitcoin investment, to address RQ2. The remainder of this section discusses the contributions of our study, its practical implications and limitations, and avenues of future research.

5.1. Seven bitcoin-specific motivations relevant to bitcoin investment

In previous research applying regret theory to explain investments in portfolios (Solnik & Zuo, 2012), currency hedging (Michenaud & Solnik, 2008) and IT (Lankton & Luft, 2008; EH Park et al., 2016), we see that different motivations are relevant to each of those contexts. A review of relevant literature reveals a gap in research explaining the drivers bitcoin investment (see Table 2), such that we cannot say which configurations of motivations are most relevant to bitcoin investment. Our research thus contributes to the question of "what" motivations are relevant in the bitcoin investment context. In our qualitative study, following the structure of regret theory (Loomes & Sugden, 1982), we show which perceptions about bitcoin, regret sentiments and personal factors are relevant.

We identify three different perceptions about bitcoin which are components of configurations of motivations that drive bitcoin investments: *profit expectancy, ease of bitcoin acquisition*, and *support of bitcoin ideology*. *Profit expectancy* is a well-known and typical motivation for investment decisions in various contexts. Among others, we know that individuals expect financial returns on stock and equity investment (Lewis & Mackenzie, 2000) and see similar expectations in crowdfunding investments, especially in equity-based crowdfunding investments (Hoegen et al., 2018). Expectations can also manifest in other forms, as in reward-based crowdfunding investments when individuals expect non-monetary returns including early access to new products or discounted price (Hoegen et al., 2018). This is similar to initial coin offerings, as expected rewards include specific services or monetary rewards (Adhami et al. 2018). We see that investments in bitcoin depend on profit expectations, which is similar to the related studied contexts of investments (see Table 2). *Ease of bitcoin acquisition* is a further motivation for investments, which has not been studied in related investment contexts (see Table 2). It is specific for bitcoin investment because it is an IS-governed investment coupled with using the Bitcoin blockchain application. This sheds new light on individual motivations to invest in bitcoin. Similarly, *support of bitcoin ideology* is specifically relevant to bitcoin investment. Bitcoin embodies a new ideology representing decentralisation, independence from intermediaries and the empowerment of individuals, such that this characteristic distinguishes bitcoin investment from other forms of investment. Among others, stock and equity investments and crowdfunding investments are governed by intermediaries, so the ideology of decentralisation is not relevant for investments decisions. Early research on investments in initial coin offerings indicates that ideology might be relevant because there is no intermediary (Fisch et al., 2019). As stock and equity investments are mediated and governed by intermediaries, those investments do not represent an ideology of decentralisation. Little is known about whether there are relevant ideology perceptions about centralised investments. Though research does not provide insights on whether the ideology influences investments, research highlights that non-economic motives are relevant. For instance, some individuals do not invest in companies they perceive to be unethical, for instance, companies with poor pollution records (Lewis & Mackenzie, 2000). Hence, we contribute by identifying a new non-economic motivation relevant for investments in the context of bitcoin.

Previous research has studied the influence of several personal factors on investing (see Table 2). Our research demonstrates that *investment skills* and *risk affinity*, grouped in configurations, influence bitcoin investment. This underscores previous research into stock and equity investment, which highlights that higher knowledge about investments leads to investments and can explain the amount invested in stocks (Balloch et al., 2015; van Rooij et al., 2011). In other words, people who invest heavily in stocks and equities tend to be highly knowledgeable about such investments (van Rooij et al., 2011). In contrast to stock and equity investments, research into crowdfunding investment supposes that the low entry barriers attract non-professional investors with low investment skills and knowledge (Hoegen et al., 2018). Further, in identifying risk affinity, our findings support previous findings in stock and equity investment indicating that risk-aversion decreases individual investment intention (Cohn et al., 1975). Bearing in mind that individuals cannot change their risk affinity (Ehm et al., 2014), we contribute that some personal factors that individuals can influence, such as investment skills, as well as some personal factors that individuals have

less influence over, such as risk affinity, influence investing in bitcoin.

Furthermore, we identify the two regret sentiments which are components of configurations of motivations that drive investing in bitcoin: *anticipated inaction regret* and *experienced inaction regret*. While some initial research into stock and equity investments focuses on how regret avoidance influences investment behaviour (Barber & Odean, 1999; Jain et al., 2015; Rau, 2015), less is known about whether such regret is anchored in the past or the future (see Table 2). This study contributes to this discussion by showing how regret can be anchored in the future (anticipated inaction regret if no investment action is taken) or in the past (experienced inaction regret that an investment action was not taken in the past).

In summary, we see that the motivations influencing bitcoin investment differ from previously studied motivations in similar contexts, such as equity and stock investment, crowdfunding investment and initial coin offering.

5.2. Four distinct configurations leading to bitcoin investment

Beyond identifying relevant motivations, we aim at explaining how they influence bitcoin investment in configurations. Taking a fsQCA approach, we reveal one necessary condition and four distinct configurations of the identified motivations driving investing in bitcoin, and we reveal the asymmetric relationship between the motivations. This answers our second research question, contributing to the theoretical "how" by explaining how configurations of the identified motivations influence bitcoin investment.

First, we contribute by revealing the importance of investment skills, which are a necessary condition of bitcoin investment (see Figure 4). We know from previous research that individuals prefer to invest in areas in which they have greater expertise (Clark-Murphy & Soutar, 2005). Our findings show that individuals will not invest in bitcoin unless they have sufficient investment skills. This underscores a characteristic of bitcoin investments that is different from related fields including investments in equity and stock, crowdfunding or initial coin offering investment (see Table 2).

Second, we contribute by revealing four configurations explaining what configurations of motivations drive investment in bitcoin. Previous investment literature has mostly focused on profit expectation as relevant motivation. Among others, research indicates that even though other aspects, such as the brand or product, are relevant for investments, individuals only invest when they expect a reward or profit (Beal et al., 2005; Bretschneider & Leimeister, 2017). Even in socially responsible investment literature and in ethical investment literature, research indicates that individuals only invest when they expect profit (Lewis & Mackenzie, 2000; McLachlan & Gardner, 2004). Our study reveals that profit expectancy is not a necessary condition of bitcoin investment, in that one configuration, the *not-profit-driven ideological investor* configuration, individuals invest in bitcoin even if they do not expect profit. In this configuration, investors are driven by their support of pro-social bitcoin ideology. The remaining three configurations include profit expectation: The *experienced ideological investor* places a high value on the ideology of bitcoin, the *risk affine profit investor*'s support of bitcoin ideology does not influence the investment, and the *incompatible profit investor* does not support the values of bitcoin.

Third, we identify an asymmetric relationship among motivations driving bitcoin investment, which means that bitcoin investment is grounded in configurations of motivations rather than isolated motivations. Within these configurations, a motivation can be present (e.g., presence of support of bitcoin ideology for the *not-profit-driven ideological investor*), absence (e.g., absence of support of bitcoin ideology for the *incompatible profit investor*) or can have no influence, meaning that they are overridden by other motivations (e.g., support of bitcoin ideology for the *risk affine profit investor*). More precisely, we find that depending on the configuration, the absence of a motivation can influence investment in terms of profit expectancy, support of bitcoin ideology, and risk affinity. Depending on the other motivations in a configuration, some motivations, in terms of ease of bitcoin acquisition, support of bitcoin ideology, risk affinity, experienced inaction regret, and anticipated and inaction regret, have no influence at all. Rather, these motivations can get overridden by other motivations in the configuration. Finally, our findings illustrate equifinality in bitcoin investment, since all four configurations consist of different motivations that all drive bitcoin investment.

5.3. Practical implications

Understanding how configurations of motivations drive bitcoin investment can help Bitcoin companies, such as wallet providers or crypto exchanges, better understand potential and existing customers, optimise their marketing, advertising and public relation activities and attract more customers. Specifically, understanding what perceptions about bitcoin, personal factors and regret sentiments motivate the four types of bitcoin investors can aid crypto exchanges in the fierce battle over customers. Our findings underscore the need to keep the bitcoin acquisition process as simple as possible to attract more individuals to invest in bitcoin. This means that wallet providers need to

design user-friendly and easily mastered wallets that enable intuitive use. Crypto exchanges should provide easy access to bitcoin, for instance, by enabling individuals to buy bitcoin using common payment methods such as PayPal or a credit card. Furthermore, as investment skills are a necessary condition for all investor types, customers lacking these skills will not invest in bitcoin. This means crypto exchanges must therefore either attract potential customers with the necessary investment skills or give potential customers the skills they need to make sound investments. For instance, crypto exchanges could provide tools to analyse price trends, inform customer regularly about price trends or give buy or sell recommendations. The results also confirm marketing and advertising strategies aimed at different target groups. This means that we reveal four different target groups, which can be differently targeted with advertising and marketing messages. For instance, companies can launch one marketing campaign to promote and support the ideology of bitcoin and a different marketing campaign to communicate the profit-making potential of bitcoin investment. A better understanding of investment in bitcoin can be used to increase the number of individuals possessing bitcoin and thereby paving the way for bitcoin to be better suitable as a means of payment (Mattke et al., 2020).

Investors aiming to understand the price dynamics of bitcoin also benefit from our results, as changes in the motivations can be a good predictor of future price changes and guide their investments. Individuals who aim to launch their own cryptocurrency to finance a venture can apply our results to the design of their own cryptocurrency. For instance, the participation in the acquisition process needs to be simple and the ideology of the venture as well as the expected profit should be clearly communicated.

5.4. Limitations and future research

In this study, we apply regret theory to explain why individuals who are aware of bitcoin, invest in bitcoin. By only sampling individuals who are aware of bitcoin, we limit the results to individuals with a certain blockchain knowledge. Individuals who are not well informed about bitcoin might have a more negative perception of bitcoin and associate bitcoin with the darknet and illegal activities. Thus, blockchain awareness as well as blockchain knowledge might shape the way individuals perceive the benefits of bitcoin and consequently point to additional motivations for investing in bitcoin. Among others, high blockchain awareness and knowledge may also imply that new beliefs arise, such as an increase in privacy needs, so that individuals might also invest in bitcoin because of their desire for more privacy. Considering the large body of research on trust in traditional IS (Lankton et al., 2015), trust might also influence blockchain investment behaviour. Specifically, (dis)trust in the blockchain application or (dis)trust in stakeholders, such as mining pools, crypto exchanges or core developer, might drive individual bitcoin investment.

From the perspective of the IS use lifecycle (Maier, 2020), we consider "non-investors" and aim to understand what motivates them to invest. Given the phases of the IS use lifecycle, future research should focus on how investment motivation differs across each phase. For example, what configurations of motivations drive continuous bitcoin investment among active investors or reinvestment among former investors. Such research could apply expectation confirmation theory to discuss the role of satisfaction on continuous bitcoin investment or reinvestment behaviour. Similarly, future research could investigate what push and pull factors motivate active investors to stop investing.

Research in other investment contexts shows that individual investment behaviour is influenced by others (Bretschneider & Leimeister, 2017). In the bitcoin context, future research should examine the role of social influence, such as peer pressure, herding behaviour, or the influence of the actions of cryptocurrency experts and gurus.

Whereas our research focuses on positive motivations of bitcoin investment, future research could investigate what demotivates bitcoin investment, building on research finding that the motivations for performing a behaviour differ from the motivations for not performing a behaviour (Liu et al., 2017). Such motivations may include negative reputation associated with scandal, legal questions, or the environmental impact of the proof of work energy consumption.

In addition, since bitcoin's ability to act as a currency depends on its capability to act as a medium of exchange, a unit of account and a store of value (Mattke et al., 2020), future research should study how the capabilities of bitcoin as a currency hinder or enable the use of bitcoin as a means of payment and what factors drive investors to start using bitcoin as a means of payment.

Finally, since profit expectancy is not a necessary condition for bitcoin investment, in contrast to most forms of investments, further research is needed into how ideological perceptions and blockchain-specific beliefs shape other non-traditional investment methods, such as initial coin offerings (ICO) or security token offerings (STO).

6. Conclusion

Bitcoin's market capitalisation has increased from zero to currently around 186 USD billion[3] in just eleven years, but little is known about the motivations that drive bitcoin investment. To fill this research gap, we

followed a mixed methods approach, conducting a qualitative study using interviews and a quantitative study using fsQCA on survey data to understand the motivations and configurations of motivations drive bitcoin investment. We analysed data collected from 76 interviews to identify motivations to identify such motivations. We then surveyed 150 individuals and derived four unique configurations of these motivations correlating to four unique types of bitcoin investors. Our results indicate that some investors invest in bitcoin because they support the bitcoin ideology. Whereas profit expectancy is not a necessary condition, some are motivated principally by profit expectations. This study lays the groundwork for future studies of the role of cryptocurrencies in society and provides valuable insights that can be used to foster the development of bitcoin and other cryptocurrencies.

Notes

1. We capitalise Bitcoin when referring to Bitcoin as an ecosystem. We refer to the Bitcoin blockchain application when talking about the protocol and the payment network. We write bitcoin with a lowercase "b" when referring to the cryptocurrency.
2. Prices based on the market price from June 2 2020.
3. Prices based on the market price from June 2 2020.

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Appendix A: The Bitcoin blockchain application

The Bitcoin blockchain application, with bitcoin as a cryptocurrency, is a public peer-to-peer payment application that stores the transaction history in a distributed ledger as a blockchain (Nakamoto 2008). A blockchain depicts a fully distributed database where all transactions are grouped into blocks. Each block has a reference to the previous block, which forms a sequential and temporal order of blocks in a chain. Therefore, the entire transaction history is stored in a specific type of distributed ledger called a blockchain (Lacity 2018; Mattke et al. 2019). The Bitcoin blockchain application is publicly accessible, which means anyone can join the Bitcoin blockchain application and read the transactions in the blockchain-ledger. The Bitcoin blockchain application uses cryptography, such as public-private keys encryption, to validate bitcoin transactions. It also uses the proof-of-work consensus mechanism, where a computation-intensive task needs to be solved to write new blocks of transactions to the blockchain-ledger (Lacity 2018; Nakamoto 2008). The proof-of-work consensus mechanism ensures that everyone participating in the Bitcoin blockchain application reaches consensus on what transactions have been completed. It is also permissionless, meaning that permission from some centralized authority is not required to participate, but, rather, anyone can participate. It provides an incentive for everyone to participate in the consensus mechanism, as successfully finish first the computation-intensive task and writing a new block of transactions to the blockchain-ledger, is rewarded with receiving a fixed number of bitcoins. Bitcoin refers to the cryptocurrency of the Bitcoin blockchain application. Bitcoin serves as a cryptocurrency, which only can be transferred within the Bitcoin blockchain application and which is limited to 21 million bitcoins (Nakamoto 2008). The incentive to get bitcoin for being the first to successfully execute the proof-of-work consensus mechanism provides economically incentive so that many participate in executing and have a low incentive to misuse the consensus mechanism (Nakamoto 2008).

Appendix B: Regret theory in previous research

Table 1 Application of Regret Theory in Operations & Information Systems

Authors	Key findings
Bleichrodt and Wakker (2015)	Introduction of a measure for regret theory which supports the basic assumptions of regret theory.
Engelbrecht-Wiggans and Katok (2008)	Higher aversion to regret overpricing results in lower average bids. Higher aversion to regret of losing auction results in higher bids.
Jiang et al. (2017)	Anticipated regret has a strong effect ton individual purchase decision, on firms' profits and product innovation.
Lankton and Luft (2008)	Based on regret theory, the paper shows that IT investment decisions are less aggressive as uncertainty increases.
Nasiry and Popescu (2012)	Regret explains two behavioural patterns for advance purchase decisions. First, an individual's delayed purchase and second, an individual's early purchase a negative surplus.
Özer and Zheng (2016)	Regret and product perceptions have a significant effect on an individual's purchase decision.
Park et al. (2016)	A process model that explains the context in which regret is triggered, the strategies used to regulate regret, and how these strategies influence the valuation, creation, exercise, and revaluation of several types of IT real options.
Solnik and Zuo (2012)	Based on regret theory the authors reveal that home bias of investors is negatively related to returns from investments.
Veeraraghavan and Debo (2011)	Regret can dampen herd behaviour.

Appendix C: Differentiation of investment in bitcoins

In the following, we provide a detailed explanation of how investing in bitcoin differs from 1) stocks and equity investment, 2) crowdfunding investments and 3) initial coin offering investments.

First, investing in bitcoin is not equal to the investment in stocks. In contrast to stock investments, investing in bitcoin has no discounted cash flow, such as dividends, and thereby mainly differ from investment in stocks (Mai et al. 2018). In contrast to investment in stocks, there exists a *"lack of short-selling or derivative instruments, meaning that it is costly to trade"* (Mai et al. 2018, 23). With the possession of most shares, individuals gain a voting right. In contrast, the possession of bitcoin does not guarantee a voting right, as 'one-CPU-one-vote' is determined in the Bitcoin blockchain application (Nakamoto 2008). Therefore, the possession of bitcoin does not give any voting right. As bitcoin are a IS governed cryptocurrency, there is no person in charge or no company managing the bitcoin cryptocurrency, which influences an individual's behaviour. In traditional investments, such as in stocks, investment depends on the perception that the persons in charge invest the necessary effort to grantee success (Hoegen et al. 2018), which again is hard if not impossible to transfer to the Bitcoin blockchain application which acts without any person in charge. Investing in stocks is not an exclusively IS-enabled investment and the investment is mediated by traditional intermediaries.

Second, investing in bitcoin is not equal to investing in crowdfunding. Crowdfunding[1] focuses on the founding of ventures of entrepreneurial individuals, who actively manage the crowdfunding campaign, and can be classified into the financing phase of a project. Crowdfunding draws on a large number of individuals, which normally contribute with small amounts, to finance a venture (Mollick 2014). Crowdfunding does not depend on the standard financial intermediaries, such as banks, however, is managed by new intermediaries, such as the crowdfunding platforms Kickstarter or Indiegogo (Hoegen et al. 2018), and thus the investment in mediated by intermediaries. Thus, in contrast to investing in bitcoin, investing in crowdfunding is about funding a non-yet realized project, which is advertised and managed by individuals on the platforms of crowdfunding intermediaries. Literature in crowdfunding furthermore indicates that the founder, as well as the presentation style of the campaign and the way of listing the crowdfunding campaign, are success factors (Vismara 2016).

[1]We refer to reward based crowdfunding and equity based crowdfunding, but not to donation based or lending-based crowdfunding, which per se is completely different from investment (Vismara 2018; Hoegen et al. 2018). For the reason of simplicity, we only refer to the term crowdfunding.

Third, investing in bitcoin is not equal to investing in initial coin offerings (ICO). Like crowdfunding, ICO raises capital to fund a venture by selling tokens through a blockchain application, which mostly have a utility function (utility tokens), an ownership right or royalties functionality (equity tokens) (Adhami et al. 2018; Fisch 2019). It shares similarities with investing in bitcoin as ICO are exclusively IS-enabled and IS-governed investments. In contrast to investing in bitcoin, investing in ICO has a clear focus of funding a venture of individuals or a company. Additionally, investing in ICO is also classified into the financing phase, thus no counter-value or real-world use exist at the time of investment.

Appendix D: Interview guideline

Table 2 Semi-structured Interview Guideline

1 Demographic Questions
What is your profession?
Is your profession related to blockchain or investment?
2 General bitcoin and investment questions
How did you become aware of bitcoin and how would you describe bitcoin?
Why would you invest in bitcoin?
What circumstances led to the decision to invest in bitcoin?
3 Different investment motivation questions
What motivations influence your bitcoin investments?
4 Follow up questions (optional)
What makes bitcoin investment attractive for you?
Why would you invest in bitcoin, and what might be the reasons why other people don't invest?

Appendix E: Overview of coding

Overview of analysis of interview data and descriptive and interpretive coding (Myers 2013).

Step 1. Descriptive coding. At the beginning of the coding approach, we descriptively coded all passages which provide insights into participants' intention to invest in bitcoin. Passages that contained multiple statements on why the participant intends to invest in bitcoin were separated into their component parts. Descriptive coding attaches descriptive labels to every applicable passage and captures the core issues expressed by the participant (Myers 2013). Whenever possible, we used the exact words of the participant to label the descriptive codes, such as 'making profit', 'belief in the system of bitcoin'. The result of the first step was a list of 32 different descriptive codes that are used as the basis for step 2.

Step 2. Categorizing into the three motivations. The goal of the second step was to categorize the identified descriptive codes into the three motivations (perceptions about bitcoin, personal factors, regret sentiments). Therefore, for each descriptive code, the interview transcript was read again to understand the context of the descriptive code and to ensure a consistent understanding of the descriptive codes across the interviews. We categorized all descriptive codes referring to perceptions about bitcoin into 'perceptions about bitcoin', all descriptive codes that refer to personal factors into 'personal factors', and all descriptive codes referring to regret sentiments into 'regret sentiments'. Three of the authors individually categorized the descriptive codes. The initial agreement score was 97.92 percent with a free-marginal multirater kappa of 0.97 (Randolph 2005; Warrens 2010), which depicts a very high level of agreement (Fleiss et al. 1981). We discussed our results afterwards to ensure consistent categorization. The result of the second step was the categorization of the descriptive codes into perceptions about bitcoin (22 descriptive codes), personal factors (4 descriptive codes) and regret sentiments (6 descriptive codes).

Step 3. Interpretive coding. In this step, we applied interpretive coding to the descriptive codes, meaning that the descriptive codes were analysed in a more structured way. The interpretive codes captured the core meaning of the descriptive codes and aggregated several descriptive codes into one interpretative code. For instance, the different descriptive codes 'more valuable in the future', 'chance of making profit' and 'chance of getting rich' were interpretatively coded as 'profit expectancy'. We identified three interpretative codes for 'perceptions about bitcoin', and for 'personal factors' and 'regret sentiments' we identified two interpretive codes each.

Step 4. Formal review of the interpretive coding. A second author reviewed the preliminary interpretive coding by repeating step 3. The initial agreement-score for the coding of 'perceptions about bitcoin' was 95.45 per cent with a free-marginal multirater kappa of 0.93. The individual coding outcomes were then compared and discussed to ensure complete agreement on the interpretive coding and thus consistent coding (Randolph 2005; Warrens 2010). The initial agreement-score for 'personal factors' and 'regret sentiments' was 100 per cent.

Table 3 Coding Overview of Study 1

Example quotes	Descriptive codes	Interpretive codes
"The chance to make profit… meaning you can easily make so much profit, that's what motivates me.", "The main reason to invest in bitcoin is clearly that I expect that bitcoin's value will increase in the next years.", "One aspect is that bitcoin can make excessive price jumps."	Profit, making money, valuable in the future, chance of making profit, secure standard of living in retirement, chance of getting rich, large price jumps, high return, good financial perspective	Profit expectancy
"One reason why I will invest in bitcoin is that it is straight forward and easy", "Investing in bitcoin is easy and uncomplicated." "I can invest even a small amount, that's what makes the investment attractive."	Easy to invest, any amount possible to invest, fraction of bitcoin possible, just buying bitcoin, easy to get wallet, easy to acquire, uncomplicated investment process, low bureaucracy for investment	Ease of bitcoin acquisition
"Investing in bitcoin is like being part of modern history. I feel this is the next big step, to cut out the middleman and form a global currency.", "I believe in the ideology of bitcoin and I trust the concept of Blockchain, which is why I would invest in Bitcoin.", "Bitcoin cuts out the middleman, and I prefer to invest in a currency which is not regulated by some corrupt organizations. "	Belief in technology, convinced of bitcoin payment, support ideology of bitcoin, support of decentralization, cut out the middleman	Support of bitcoin ideology
"I have a good feeling for investments and I mostly know where to invest. So, I will invest in bitcoin.", "I think I have everything it takes to make good investment decisions; therefore, I definitely will invest in bitcoin." "In the last years, it has proven that I got a good sense to do the right investments, so that motivates me to invest."	A good feeling for investments	Investment skill
"I am willing to take the risk on investments, so that's definitely why I invest. I think taking risk is normal for such an investment", "I am the type of guy who sometimes takes a risk and that is the same with the decision to invest in bitcoin. I see the risk as normal for such an investment.""Because I am willing to take a risk with this investment."	Acceptance of risk, preference for risky investments, willing to take risk	Risk affinity
"I was thinking about investing in bitcoin in early 2011 and did not take the chance. That's why I want to invest now.", "I was confronted with the decision whether to invest in Bitcoin in early 2015 and I decided not to invest. I could have been millionaire today, which is what I regret. At least I will invest in bitcoin now.", "I could have been rich, if I invested earlier, that motivates me to invest now."	Regret not investing, feeling sorry for not investing, could have been rich	Experienced inaction regret
"I do not want to miss the chance now, or I will regret the decision in the future.", "I truly believe that I would one day regret my decision to not invest in Bitcoin, and I don't want to regret that. So, I think I will invest some of my money in Bitcoin.", "Don't want to feel the regret if the price increases, so I jump on the bandwagon."	Will regret not investing, regret if price increases further, regret if it is more valuable in the future	Anticipated inaction regret

Appendix F: Descriptive statistics and common method bias

As recommended (Podsakoff et al. 2003) for self-reported data, we additionally tested for common method bias (CMB) with two tests. We applied Harman's single factor test, which shows that only 41 per cent of the variance of the data is explained by one factor, and we examined the correlation matrix (Table 4) for extremely high correlations (r > 0.90), as specified by Pavlou et al. (2007). These results reveal that CMB is not an issue in this study.

Table 4 Descriptive Statistics and Discriminant Validity

		Mean	SD	AVE	CR	α	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
(1)	Support of bitcoin ideology	4.18	1.43	0.76	0.93	0.91	0.87							
(2)	Anticipated inaction regret	4.37	1.52	0.80	0.89	0.76	0.09	0.90						
(3)	Experienced inaction regret	4.79	1.75	0.89	0.94	0.88	-0.03	0.47	0.94					
(4)	Ease of bitcoin acquisition	4.94	1.3	0.87	0.93	0.85	0.35	0.26	0.18	0.93				
(5)	Intention to invest in bitcoin	4.63	1.57	0.92	0.96	0.91	0.07	0.69	0.48	0.30	0.96			
(6)	Profit expectancies	4.67	1.26	0.70	0.92	0.89	0.27	0.57	0.33	0.44	0.60	0.84		
(7)	Risk affinity	4.72	1.15	0.68	0.89	0.84	0.44	0.38	0.28	0.49	0.44	0.48	0.82	
(8)	Investment skills	5.02	1.25	0.73	0.92	0.88	0.17	0.40	0.52	0.60	0.61	0.54	0.56	0.85

Note: square root of AVE is listed on the diagonal of bivariate correlations; SD = standard deviation; AVE = average variance extracted; CR = composite reliability; α = Cronbach's alpha;

Appendix G: Measures

The constructs used in study 2 were based on previous research and used items that had been validated in previous research. The correlation matrix includes descriptive statistics of the constructs.

Table 5 Information on Construct Measures

	Information on Construct Measures	Loading
Intention to invest in bitcoin adapted from East (1993)	I will invest in bitcoin.	0.9570.956
	I intend to invest in bitcoin.	
Profit expectancy adapted from Ali (2011)	Bitcoin is financially sound.	0.827
	Investing in bitcoin seems to be able to generate high returns.	0.781
	I believe bitcoin will perform satisfactorily in the future.	0.885
	Bitcoin has sufficient resources to grow in the future.	0.896
	I think investing in bitcoin is highly rewarding.	0.815
Ease of bitcoin acquisition adapted from Pavlou and Fygenson (2006)	Purchasing bitcoin online would be easy.	0.875
	Learning how to purchase bitcoin online would be easy.	0.852
Support of bitcoin ideology adapted from Karahanna et al. (2006)	Investing in bitcoin runs counter to my own values.	0.812
	Investing in bitcoin does not fit the way I view the world.	0.851
	Investing in bitcoin goes against what I believe blockchain should be used for.	0.949
	Investing in bitcoin is not appropriate for a person with my values regarding the role of blockchain.	0.875
Investment skills adapted from Pavlou and Fygenson (2006)	If I wanted to, I could become skilful at making good investment decisions about bitcoin.	0.892
	If I wanted to, I could easily become knowledgeable about investing in bitcoin.	0.851
Risk affinity adapted from Allen et al. (2005)	I am willing to take risks when choosing an investment.	0.824
	I prefer a high-risk investment with a large profit over an investment with low risk and medium profit.	0.865
	I prefer investments that have high risks.	0.809
	Risk is a normal part of an investment.	0.789
Anticipated inaction regret adapted from Inman and Zeelenberg (2002) and Tsiros (2008)	If I don't invest in bitcoin now, I will regret it later.	0.848
	How much will you regret your decision if you do not invest in bitcoin now? (anchored by not regret at all/regret very much)	0.925
Experienced inaction regret adapted from Inman and Zeelenberg (2002) and Tsiros (2008)	If you could make your decision again, would you change your decision?	0.889
	How much happier would you have been if you had made a different decision? (anchored by not much happier/much happier)	0.923

Note: All items were measured with a 7-point Likert-type agreement scale ranging from 1 = 'completely disagree' to 7 = 'completely agree', if not indicated otherwise.

Appendix H: Necessary conditions

The analysis of necessary conditions reveals whether the presence or absence of a condition is necessary for individuals to invest in bitcoin. The analysis reveals that investment skills are the only necessary condition.

Table 6 Necessary Conditions

Necessary Conditions	Consistency	Coverage
Profit expectancy	0.87	0.86
~ Profit expectancy	0.41	0.69
Ease of bitcoin acquisition	0.86	0.79
~ Ease of bitcoin acquisition	0.41	0.77
Support of bitcoin ideology	0.54	0.78
~ Support of bitcoin ideology	0.74	0.79
Risk affinity	0.85	0.85
~ Risk affinity	0.46	0.73
Investment skills	**0.93**	**0.81**
~ Investment skills	0.35	0.72
Experienced inaction regret	0.85	0.83
~ Experienced inaction regret	0.41	0.68
Anticipated inaction regret	0.82	0.89
~ Anticipated inaction regret	0.46	0.66

Note: ~ indicates the absence of a condition or outcome; necessity consistency threshold = 0.90; conditions exceeding the consistency threshold are marked in boldface